

Vibe Coding: The Hype, the Reality, and the Future

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New era of AI

The world was changed!
AI can do everything!?



Oh....
Really?

Everywhere AI,
Ubiquitous AI agents

Anyone Can Code.
Anything Can Be Built.



Oh...
Cool...

New term, "Vibe Coding," has emerged.



Motivations (2/2)

Why Professor Noh has not been uploading videos to YouTube?



In fact,
I was incredibly busy.



Shock in that AI is advancing
at a breaking speed.

Now, I want to share my feeling & insight

Trends in AI-Technology

1. Competition of Algorithm Enhancement



2. The Scaling Race (LLM Size Wars)

3. Multimodal (Domain Confusion)



4. AI Agents (Ubiquity of AI, Less Human)



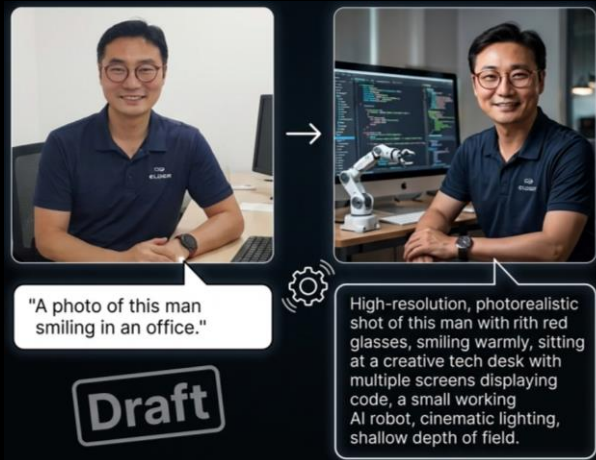
5. Physical AI (Robotics, Embodied AI)

The Next Frontier? Who knows?

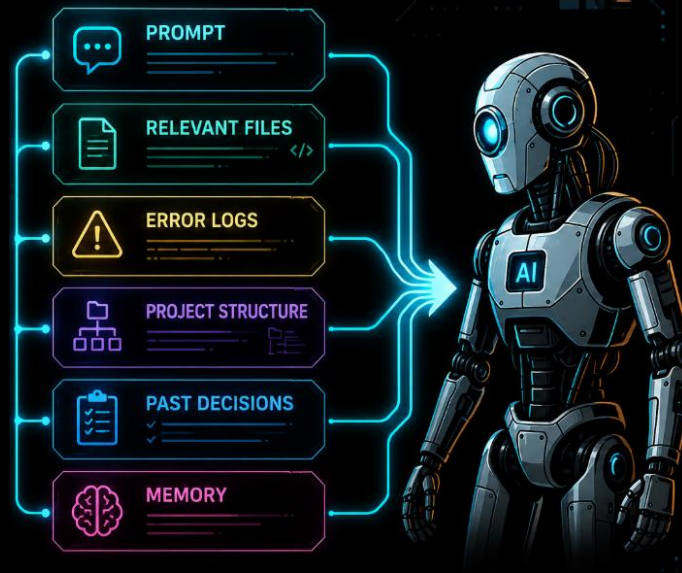
LLM Advancement: The war on AI hallucinations

Prompt Engineering

How to ask!



Context Engineering What to be shown!



In the future?
Nobody knows!



Harness Engineering AI Guardrails & Red Teaming!



AGENTS.md
CLAUDE.md
⋮

Everything Has Changed!

My thought was...

- But how is it actually implemented?
- Do We Just Plug and Play?
- Can AI replace fundamental knowledge?



But LLM-based agents
have broken All the Rules.

How to tell others?



Lights & Shadows

Every developer will
become unemployed,
and computer science
will disappear.

Vibe Coding



Super fast!

Super easy!

Everyone can do coding!

Everyone can be a SW developer!

Vibe Coding?

Andrej Karpathy



The term was coined in February 2025 by computer scientist Andrej Karpathy, a co-founder of OpenAI and former AI leader at Tesla. Merriam-Webster listed the term in March 2025 as a "slang & trending" expression. It was named the Collins English Dictionary Word of the Year for 2025.

From online wiki: https://en.wikipedia.org/wiki/Vibe_coding

The Hype vs. Reality of Vibe Coding



Note:

특정 콘텐츠를
공격하려는 의도는
전혀 없습니다.

Ultimately,

Vibe Coding Does **Not Eliminate** the Need
for Computer Science Knowledge.



str = "Please don't misunderstand me."

But! Hey, professor!

There is **no doubt** that software development
has become **easier** and far more **accessible**.



Calm down, and
hear me out please.



Simple Web Apps Are Easy with Vibe Coding



개인 블로그



딸깍



I agree with that as well.



투두 앱



회사 홈페이지



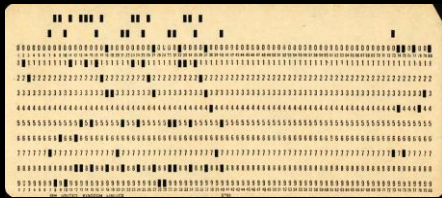
유튜브 웹페이지

Even as a professor, I use vibe coding.

Even I, as a computer science professor, use vibe coding.

The paradigm of coding has changed.

Even I, once strict and traditional, have changed.



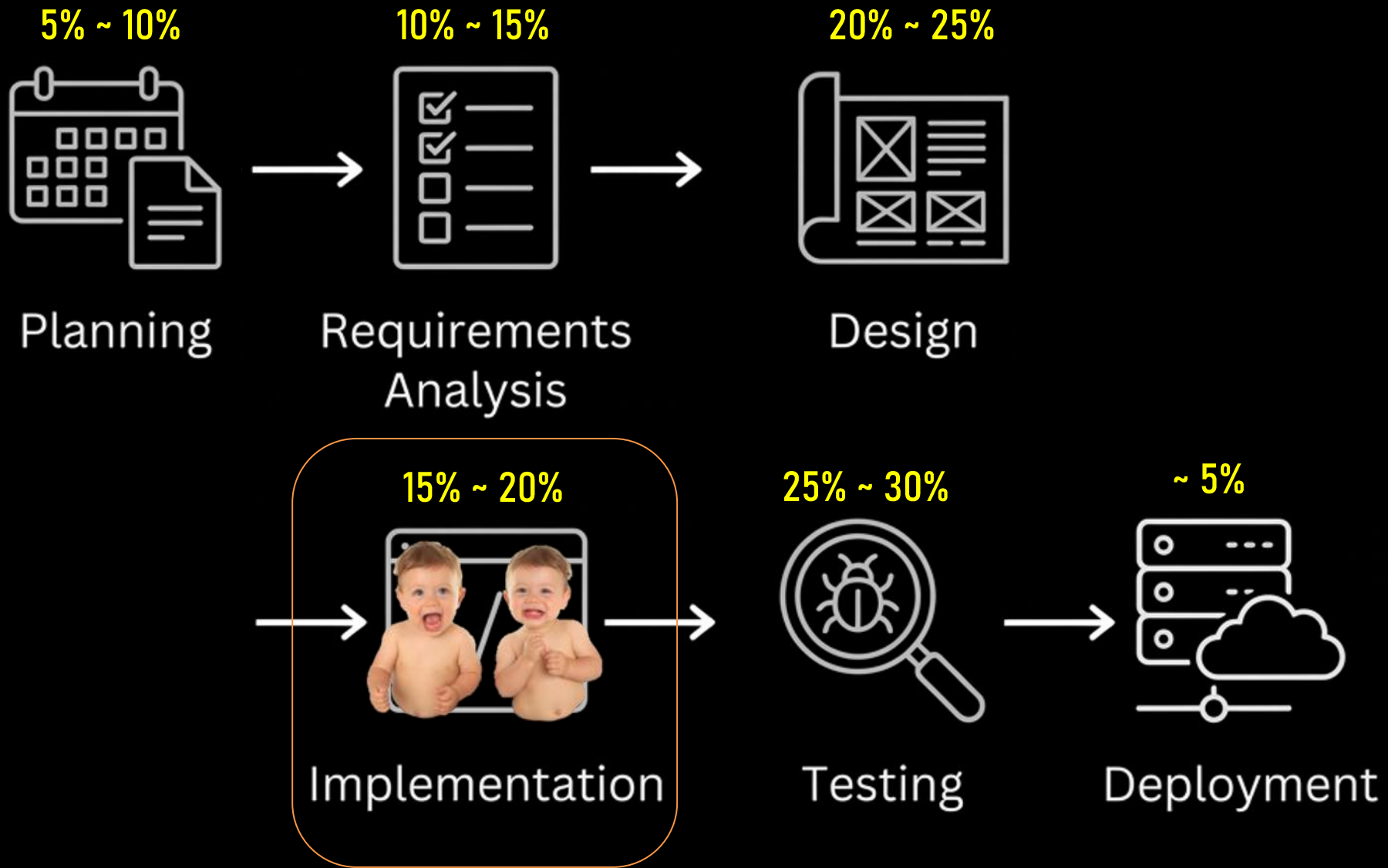
이미지 출처: 나무위키
<https://namu.wiki/w/천공카드>



Dev Language: Low (Assembly) → High (C++/Java/Python)

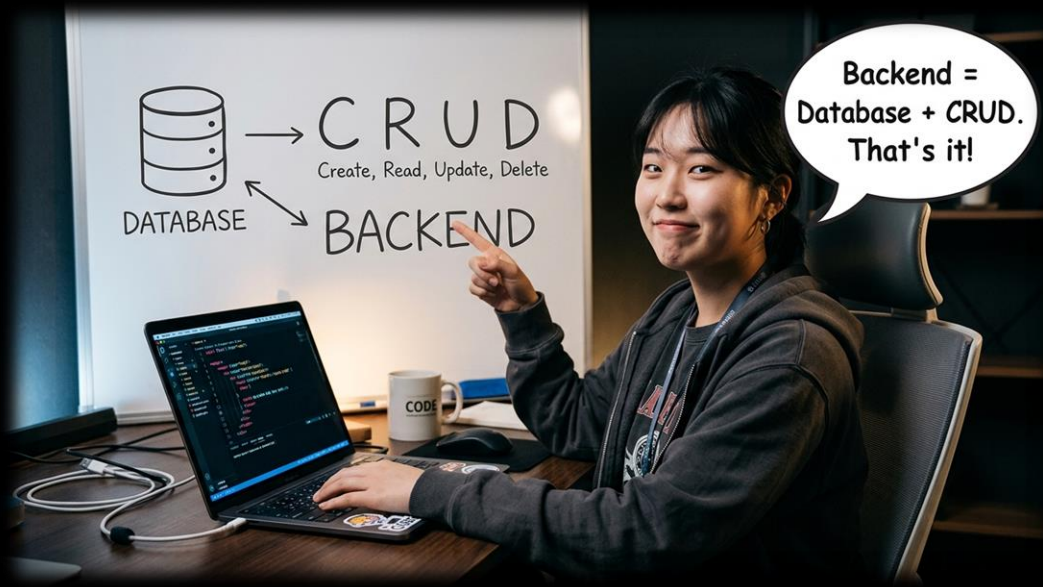
There is no reason
why vibe coding
shouldn't work.

Actually, SW Development is NOT implementation!



Misnderstanding the Role of the Backend

What about Process Logic, Network, DB,
Security, Optimization, Middleware, and etc..?



Is Vibe Coding Really Software Development?

Looks great, beautiful, and perfect!

But really?



Debugging
Nightmare!

Copy-and-Paste
Hell

Token Burnout
(Excessive API Costs)

Lack of Security
Awareness

No
Consideration
for Scalability

Limited Understanding
of Network Integration

Difficulty in
Verification and
Hallucination
Detection

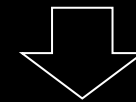
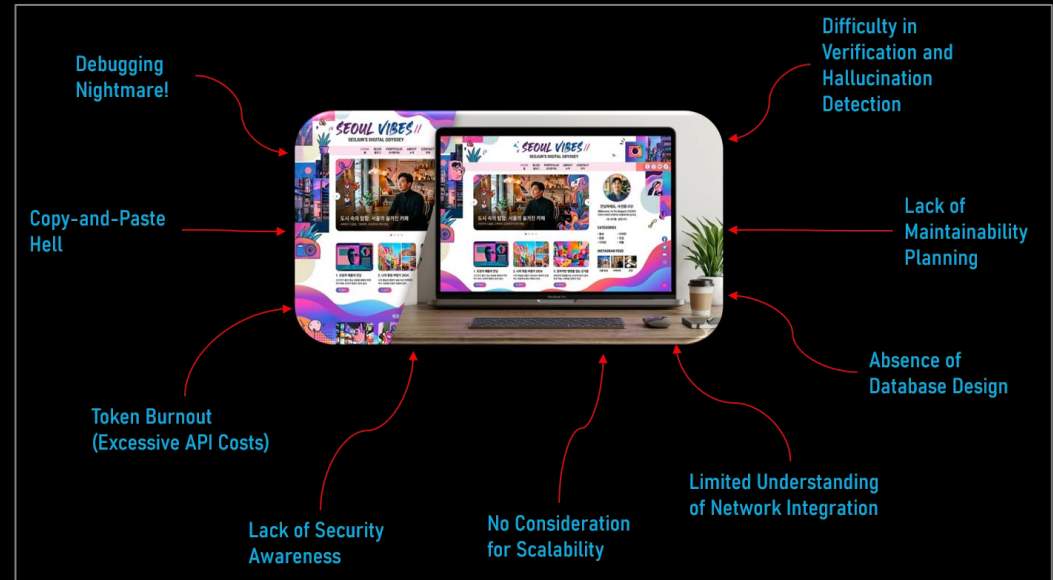
Lack of
Maintainability
Planning

Absence of
Database Design

The Uncomfortable Truth About Vibe Coding

We must have:

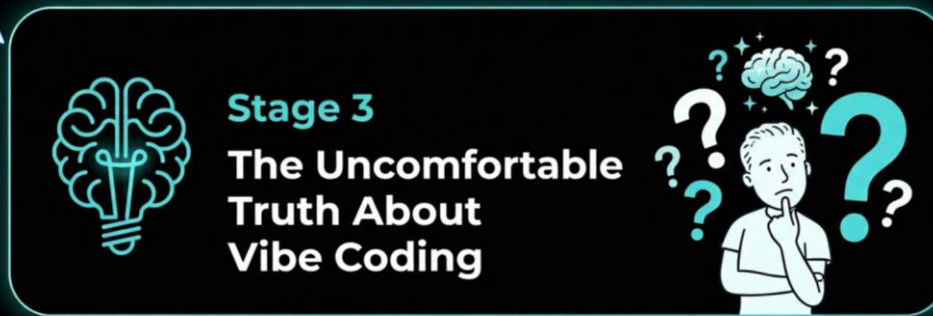
- Full controls on SW
- Availability on V&V
- Understanding about:
 - Logic Flow
 - Data Position & Structure
 - Potential Chock Points
 - Test Guard Rails



We Promise that
Built in Just 10 Seconds
What ever You Want!



In the End, We're Back to the Basics.



When should you use vibe coding anyway ?

Fast build Prototypes



Develop Utilities & Job Automation Tools



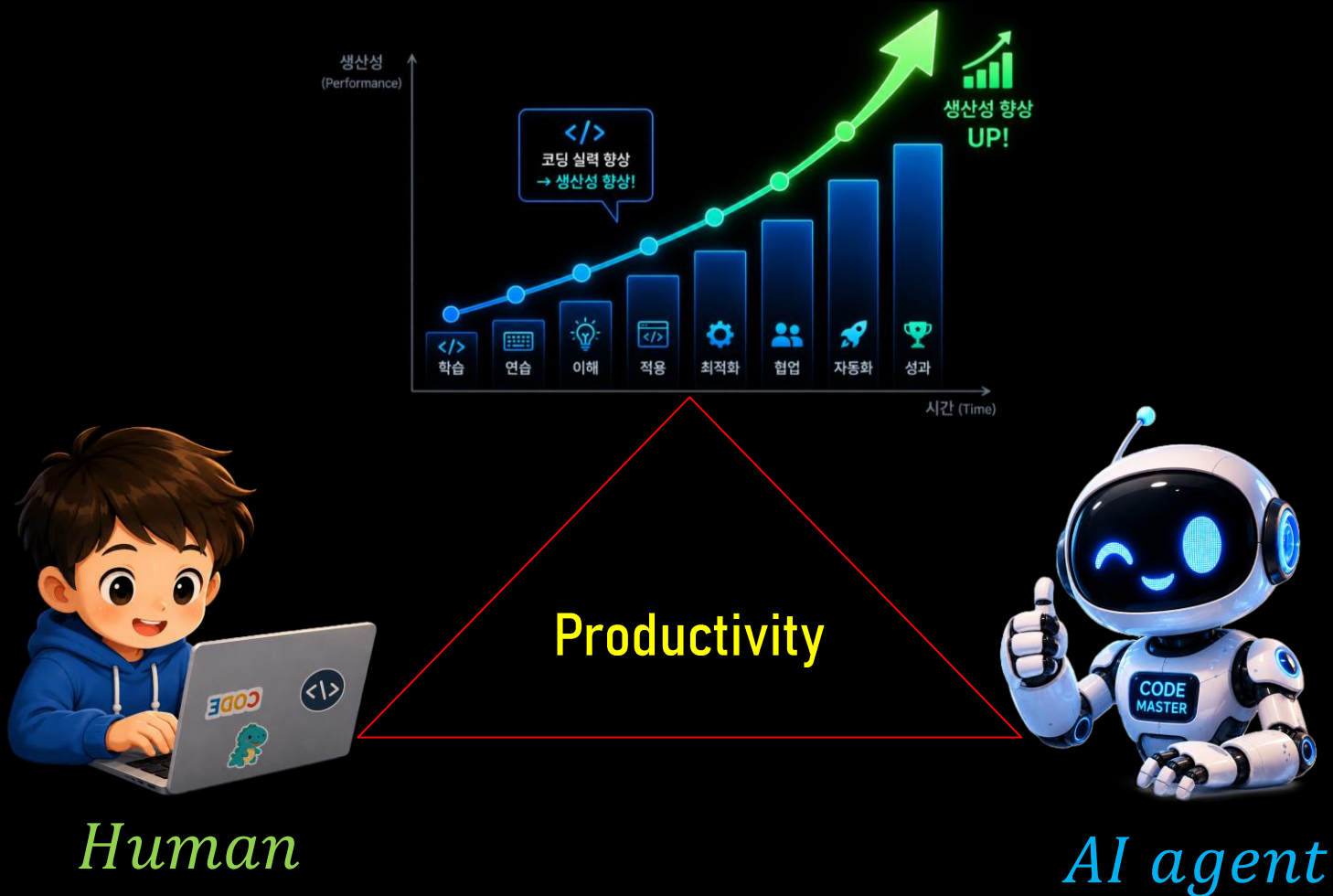
Verify MVP in 10 Seconds



SW That Tolerates Bugs



How Powerful Vibe Coding Can Be



Plus, Multiplication, Exponent



Productivity Equation

- P : Productivity
- H : Human Ability
- A : The Power of AI
- e : Natural Constant



$$P = H + A$$



$$P = H \cdot A$$



$$P = H \cdot e^A$$





수고하셨습니다 ..^^..